

CATS: Collective and Temporal Synergy for Timely Model Updating in Federated Continual Learning

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Abstract—Federated continual learning (FCL) enables a global model to be incrementally updated using distributed and sequentially arriving data without centralizing raw samples. This capability is increasingly important for Internet of things applications, such as intelligent transportation, industrial monitoring, and unmanned systems, where edge devices continuously generate or collect non-stationary data streams in dynamic environments. A key challenge is to promptly incorporate newly arrived data while preserving previously acquired knowledge under stringent resource and labeling constraints at the edge. This article presents CATS, a device scheduling framework based on Collective and Temporal Synergy for timely FCL at the network edge. CATS evaluates the learning value of distributed data by jointly accounting for temporal drift across sequential data arrivals and collective complementarity among devices. It then schedules the most valuable set of devices under resource constraints to accelerate model convergence. We further extend CATS to a semi-supervised setting, resulting in CATS-Semi, which exploits abundant unlabeled data by balancing the quantity and reliability of pseudo-labels for both newly arrived and historical samples. In this way, CATS-Semi improves label efficiency while mitigating pseudo-label error accumulation and knowledge forgetting. Two case studies demonstrate the effectiveness of CATS and highlight its potential to support timely and resource-efficient model evolution at the network edge.

Index Terms—Federated continual learning, device scheduling, temporal drift, semi-supervised learning.

I. INTRODUCTION

The convergence of machine learning and wireless networking has accelerated the development of Internet of things and connected intelligence [1]. Smart devices, such as sensors, industrial terminals, connected vehicles, and mobile robots, continuously sense their physical environments and generate data streams. By learning from these data, intelligent models can support a wide range of applications, including environmental perception, anomaly detection, industrial monitoring, and autonomous decision-making.

Traditional centralized learning typically requires raw data to be uploaded to a central server, potentially incurring substantial communication overhead, service latency, and privacy risks. Federated learning (FL) provides a promising framework for distributed model training, in which devices train models

using their local data and upload only model updates for global aggregation [2], [3]. In many practical edge environments, however, data are not collected only once. Instead, devices continuously encounter new scenarios and collect samples whose distributions evolve over time. The global model must therefore continually learn from evolving data streams while retaining knowledge acquired from previously encountered data, motivating federated continual learning (FCL) [4]. For example, as road, weather, and traffic conditions evolve across locations and time, intelligent connected vehicles can collaboratively update a shared perception model using newly collected data while preserving knowledge of previously encountered driving conditions.

In this context, the timeliness of model updating is particularly important [5]. If the global model is not updated promptly, it may lag behind the current data distribution and fail to acquire newly emerging knowledge. Conversely, overemphasizing newly arrived data may overwrite previously learned representations and lead to catastrophic forgetting [6]. The objective of timely FCL is therefore to rapidly adapt the global model to newly arrived data under limited resources while preserving knowledge learned from historical data.

Achieving this objective in wireless edge environments faces three key challenges. First, device data are spatio-temporally heterogeneous, meaning that data distributions vary across devices and evolve over time. Second, limited communication, computation, and energy resources make it impractical to involve all devices in every training round. The system needs to schedule a subset of devices that can provide the highest learning value within the available resource budget. Third, continuously arriving data are often unlabeled, while unreliable pseudo-labels may cause errors to accumulate over learning stages [7]. Therefore, timely model updating requires effectively identifying and exploiting the most valuable distributed data under resource and labeling constraints.

In this article, we propose CATS, a scheduling framework based on collective and temporal synergy for timely FCL under resource and labeling constraints. CATS balances temporal drift in sequential data, which captures the evolution of data distribution over time, with the collective divergence across device data distributions, which characterizes cross-device data complementarity and global representativeness. By leveraging this temporal-collective synergy, CATS selects resource-efficient subsets of devices that accelerate convergence while preserving previously acquired knowledge. Furthermore, we extend CATS to the semi-supervised FCL setting, yielding CATS-Semi. It exploits unlabeled data by balancing pseudo-

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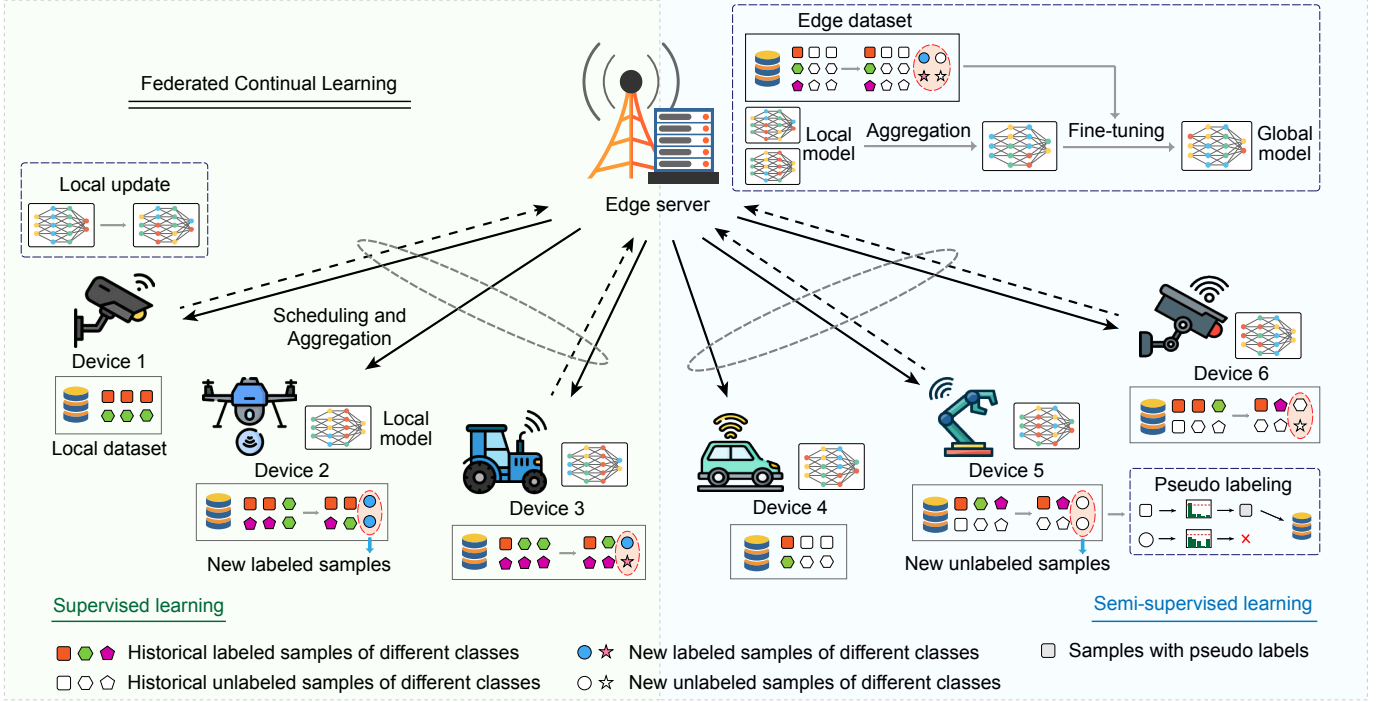


Fig. 1. Illustration of a federated continual learning system over the edge of the wireless network.

label quantity and reliability across newly arrived and historical samples, thereby improving label efficiency while mitigating pseudo-label error accumulation and knowledge forgetting.

In the following, we first introduce the system model and objective of FCL, followed by a discussion of the key challenges. We then present the CATS framework from three perspectives: learning from spatio-temporally heterogeneous data, exploiting abundant unlabeled data, and performing resource-aware device scheduling over wireless networks. Finally, two case studies demonstrate the effectiveness of CATS and CATS-Semi in achieving timely model updating and label-efficient FCL, respectively.

II. TIMELY FCL AT THE WIRELESS EDGE: SYSTEM MODEL, OBJECTIVE, AND KEY CHALLENGES

As shown in Fig. 1, devices such as vehicles, robots, and sensors continuously collect streaming data and use the incrementally arriving samples to update their local models at the wireless edge. The local data distribution of each device may evolve over time as its operating state and surrounding environment change. Such evolution may lead to a class-incremental setting, in which previously unseen classes emerge over time, for example, when autonomous vehicles encounter new types of road users or traffic signs during operation; or a domain-incremental setting, in which the label space remains unchanged while the data distribution shifts across environments, operating conditions, or time periods, for example, when the same road-object categories are observed under different weather, illumination, road, or camera conditions. The resulting local model updates are aggregated under the coordination of an edge server to jointly train a shared global model. Depending on annotation availability, newly collected

samples may be labeled or unlabeled, giving rise to a semi-supervised learning setting in which most device-side samples are unlabeled and the edge server maintains only a small labeled reference dataset for model initialization, calibration, and performance evaluation.

Timely FCL aims to maximize model performance under a given latency budget or, alternatively, minimize the overall adaptation latency required to reach a target performance level. Unlike conventional FL, the target learning performance of FCL should jointly reflect adaptation to newly arrived data and retention of previously acquired knowledge. The overall adaptation latency is defined as the elapsed time from the arrival of new data until the global model reaches a target performance. In each training round, the per-round latency is dominated by the local computation and uplink model-transmission delays of the slowest scheduled device. Therefore, the overall adaptation latency depends jointly on the number of rounds required to reach the target performance and the per-round latency.

Spatio-temporal data heterogeneity, label scarcity, and limited edge resources make device scheduling essential for timely FCL. The system must select devices that jointly provide high learning value while satisfying communication and computation constraints, thereby enabling reliable model updates within the delay budget.

A. Spatio-Temporal Data Heterogeneity

A fundamental characteristic of edge environments is the spatio-temporal heterogeneity of device data. Due to differences in deployment locations, sensing regions, user behaviors, and task states, local data distributions may vary substantially across devices and deviate from the global distribution, resulting in spatial data heterogeneity. Meanwhile, local data

distributions may evolve over time as its operating state and surrounding environment change. For example, new scenarios, objects, or classes may emerge in continuously arriving data streams, resulting in temporal data drift. Unlike standard FL formulations that assume fixed non-i.i.d. datasets, FCL must *jointly address cross-device data heterogeneity and temporal distribution drift*. Device scheduling should therefore balance the temporal value of newly arrived data with the collective representativeness of selected devices to enable rapid adaptation without catastrophic forgetting.

B. Label Scarcity and Pseudo-Label Reliability

In dynamic edge networks, devices continuously generate large volumes of unlabeled streaming data, while ground-truth labels are scarce. The small labeled reference dataset at the edge server is insufficient to represent evolving distributions and newly emerging classes. Semi-supervised strategies such as pseudo-labeling can exploit unlabeled data, but pseudo-label reliability depends on the current model and may be unreliable for newly arrived or underrepresented data. Incorrect pseudo-labels can be reinforced during local training and propagated through aggregation, leading to confirmation bias and error accumulation. Device scheduling should therefore consider the *quantity and reliability of pseudo-labeled data*, favoring devices that provide informative and correctly-labeled samples.

C. Communication and Computation Constraints

Distributed devices operate under limited and time-varying wireless bandwidth and computing capabilities. Dynamic channels introduce unstable communication delays, while heterogeneous computation leads to straggler effects, where the slowest device dominates the per-round latency under synchronous aggregation. These constraints create a trade-off between learning value and adaptation latency. Scheduling more devices may accelerate model convergence but increases delay, whereas selecting only fast devices may reduce delay but limit learning performance. Device scheduling should therefore jointly consider learning value and costs.

III. CATS: COLLECTIVE AND TEMPORAL SYNERGY FOR RESOURCE-AWARE DEVICE SCHEDULING

To address the above challenges, we propose CATS, a device scheduling framework for timely FCL that jointly evaluates the learning value and resource feasibility of candidate devices. Specifically, CATS characterizes the value of spatio-temporally heterogeneous data by jointly considering temporal drift and collective divergence. We further extend this learning-value characterization to semi-supervised FCL by balancing the quantity and reliability of pseudo-labels, yielding CATS-Semi. These learning-value measures are then integrated with device computing capabilities and wireless communication conditions to select device groups that can provide representative, reliable, and timely model updates within the prescribed latency budget.

A. Learning from Spatio-Temporally Heterogeneous Data

In conventional FL, device scheduling under data heterogeneity is often designed from an individual-device perspective. It typically prioritizes devices with smaller gradient divergence, defined as the discrepancy between the local and global gradients, while scheduling as many devices as possible under resource constraints. Our recent study shows that data heterogeneity is better characterized as a collective property of the scheduled device group rather than an independent property of each device [8]. Even when individual devices have biased local data distributions, combining devices with complementary data can make their aggregate data distribution closer to the global distribution, thereby improving the representativeness of aggregated updates and accelerating model convergence. Based on this observation, we introduce *collective divergence* to quantify the discrepancy between the aggregate data distribution of the scheduled devices and the global data distribution, and use it to guide device scheduling. Specifically, collective divergence is defined in the gradient space as the discrepancy between the weighted average gradient of the scheduled device group and the gradient over the global dataset. For classification tasks, this quantity can be further upper bounded by the weighted discrepancy between the aggregate class distribution of the scheduled devices and the global class distribution, where the distributional discrepancy of each class is weighted by its corresponding class-specific gradient norm.

Furthermore, FCL also needs to account for the temporal evolution of local data distributions. We incorporate the replay-based idea of continual learning, where each device retains a limited set of representative historical samples and trains on them together with newly arrived data. To enable timely adaptation to newly emerging classes and features, the scheduler should prioritize devices whose newly arrived data exhibit substantial distributional changes, while retaining sufficient historical data to mitigate catastrophic forgetting. We therefore introduce *temporal drift* to quantify the discrepancy between current and historical data distributions of each device [9]. Specifically, temporal drift is defined in the gradient space as the discrepancy between the gradients induced by the current and historical local data of each device. For classification tasks, this quantity can be further characterized according to the form of data evolution. In a class-incremental setting, temporal drift can be upper bounded by the weighted changes in class proportions between consecutive stages, where the historical proportion of a newly emerging class is set to zero. In a domain-incremental setting, the class proportions remain unchanged, and temporal drift can be characterized by the differences in the average model updates induced by the same classes across consecutive stages. The proposed CATS framework, short for collective and temporal synergy, employs a time-varying scheduling objective that rewards high temporal drift while penalizing high collective divergence, thereby balancing rapid adaptation with global representativeness. Early in training, CATS places greater weight on temporal drift to accelerate adaptation to newly arrived data. As training proceeds, it gradually shifts the weight toward collective

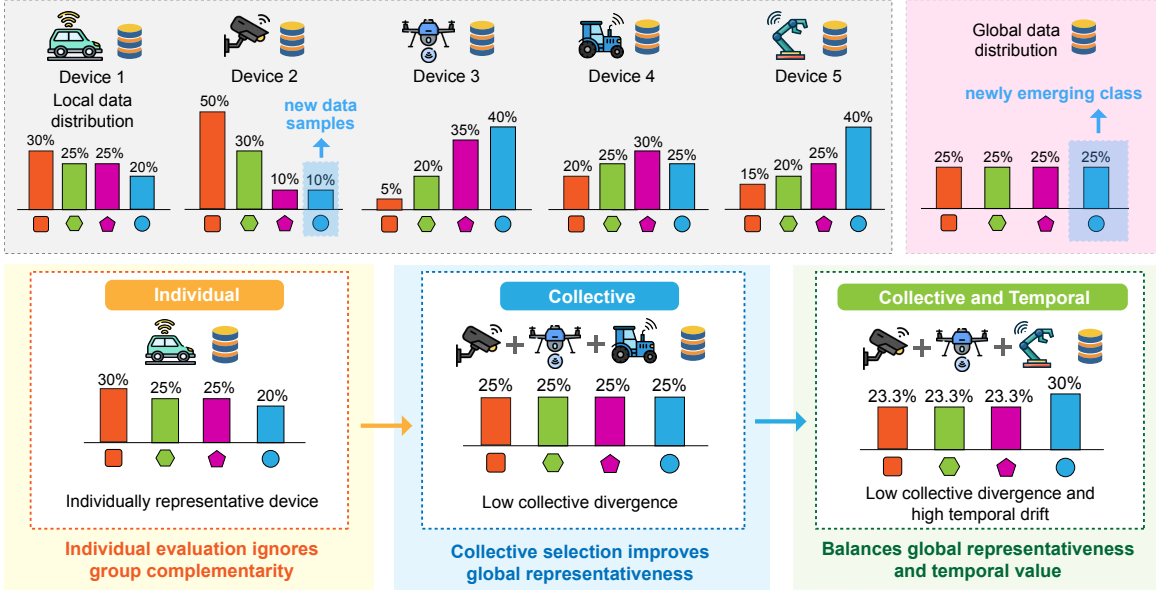


Fig. 2. Conceptual illustration of device selection under spatio-temporal data heterogeneity. CATS jointly considers device-level temporal drift and group-level collective divergence for device scheduling.

divergence, improving the representativeness of the selected device group and stabilizing performance across newly arrived and historical data. A conceptual illustration is provided in Fig. 2.

B. Learning from Unlabeled Data

In practical FCL systems, the edge server may be able to collect a small number of labeled data samples, whereas the large volume of data continuously collected by devices cannot be labeled promptly. To effectively exploit these unlabeled data, we extend CATS with semi-supervised learning, resulting in CATS-Semi. The workflow is illustrated in Fig. 3. The edge server first initializes the global model using the limited labeled reference data and broadcasts it to the distributed devices. In each round, each candidate device uses the received global model to generate pseudo-labels for its local unlabeled samples. However, pseudo-labels do not always provide reliable supervision. Incorrect pseudo-labels may introduce erroneous supervision, reinforce existing prediction errors, and eventually cause confirmation bias. Each device therefore applies a pseudo-label filtering mechanism so that only sufficiently reliable samples are used for local training. Building on collective divergence and temporal drift, the edge server further *balances the quantity and reliability of pseudo-labels* when making scheduling decisions. Here, pseudo-label quantity is defined as the number of unlabeled samples passing the confidence threshold, while pseudo-label reliability is measured by the average confidence of pseudo-labels for new-class samples. A larger pseudo-label quantity allows more new-class samples to participate in local training and helps reduce sampling uncertainty, whereas higher pseudo-label reliability helps reduce the influence of incorrect pseudo-labels and improves the accuracy of learning newly introduced classes. Finally, the server aggregates the received updates

and further fine-tunes or calibrates the global model using the labeled reference dataset.

Moreover, newly introduced and previously learned classes often exhibit different pseudo-labeling characteristics. For newly introduced classes, the model has not yet learned sufficiently discriminative representations. An overly high confidence threshold may therefore filter out many potentially valuable samples and slow adaptation to new knowledge. To address this issue, CATS-Semi initially adopts a relatively relaxed filtering criterion for newly introduced classes to retain sufficient effective pseudo-labeled samples. As the model progressively acquires more discriminative representations and its predictions become more reliable, the confidence threshold is gradually increased to suppress incorrect pseudo-labels and prevent error accumulation. In contrast, predictions for previously learned classes are generally more stable, allowing a stricter filtering criterion to be applied throughout training. This *adaptive thresholding strategy* enables CATS-Semi to accelerate early adaptation while progressively improving pseudo-label reliability.

C. Joint Scheduling under Wireless Resource Constraints

The preceding two subsections characterize device learning value from the perspectives of spatio-temporal data heterogeneity and unlabeled-data exploitation. However, devices with high learning value may not always be feasible for the current training round. Because edge devices differ in computing capabilities and wireless channel conditions, device scheduling must also account for resource feasibility. CATS therefore jointly considers temporal drift, collective divergence, and, in CATS-Semi, pseudo-label quantity and reliability, together with device computation and communication costs.

In synchronous FL, the per-round latency is dominated by the slowest device, consisting of both local computation delay

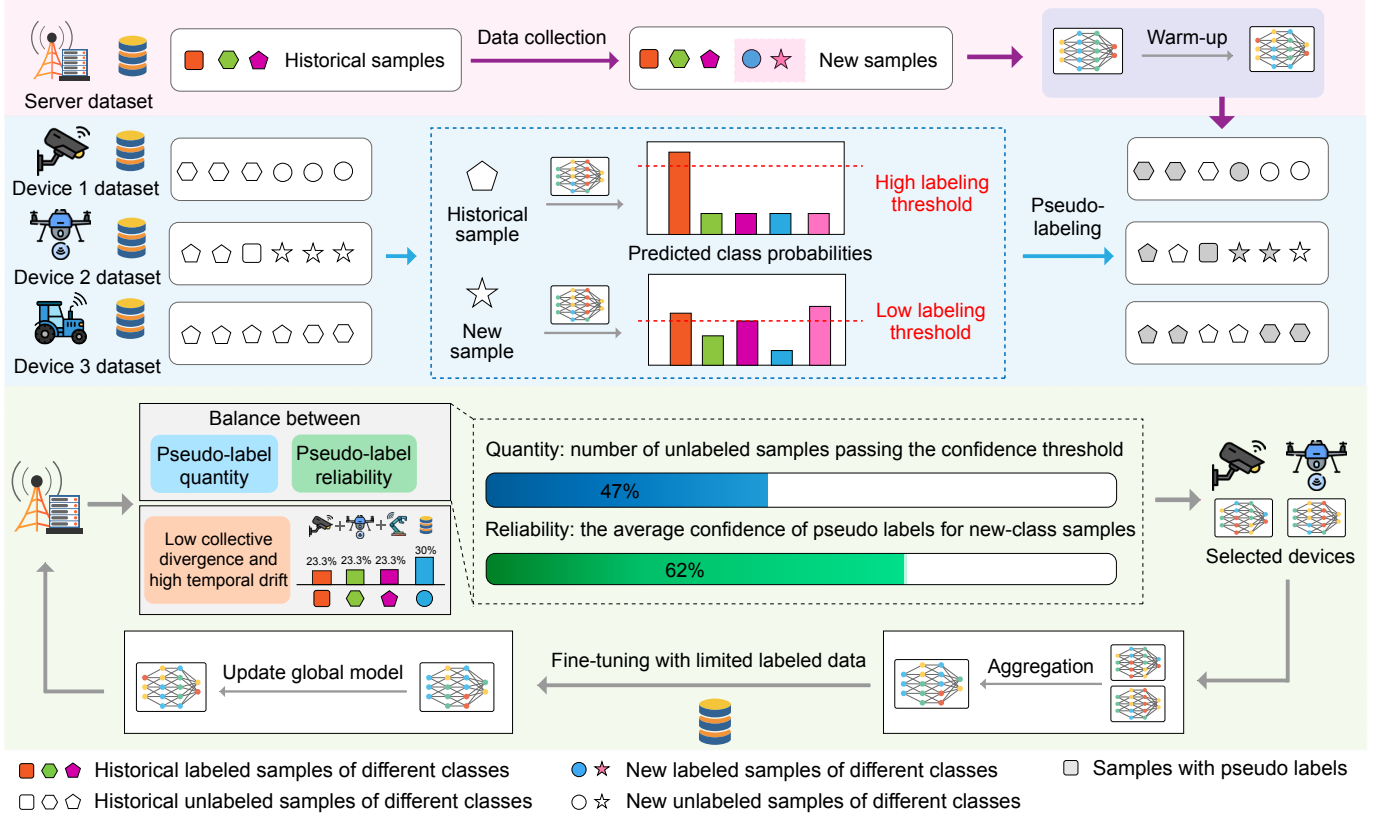


Fig. 3. Workflow of CATS-Semi for device scheduling based on pseudo-label quantity and reliability.

and uplink communication delay. Local computation latency depends on device computing capability, energy state, and training workload, whereas uplink communication latency is affected by wireless channel conditions, bandwidth allocation, and model-update size. Consequently, a device with valuable data may still be infeasible under a given latency budget because of limited computing capability or poor channel conditions. Conversely, a resource-efficient device may not be preferred if its data provide limited learning value.

This creates a fundamental tradeoff between per-round learning gain and training latency [5]. Scheduling more devices with complementary data may improve the representativeness and informativeness of aggregated updates, thereby reducing the number of rounds required to reach the target performance. However, involving more devices may also introduce stragglers, intensify resource competition, and increase per-round latency. Timely model updating therefore requires jointly controlling both the number of rounds to convergence and the latency of each round, rather than optimizing either learning gain or per-round latency alone.

Accordingly, CATS jointly evaluates the learning value and resource feasibility of candidate device groups in each training round. In CATS, learning value is characterized by temporal drift and collective divergence, while CATS-Semi further incorporates pseudo-label quantity and reliability. The edge server combines these indicators with local computation and uplink communication latency to select a high-value feasible device group under the prescribed per-round latency

budget. By improving the effectiveness of each model update while controlling per-round latency, CATS reduces the overall adaptation latency required to reach the target performance.

IV. CASE STUDY I: LEARNING FROM SPATIO-TEMPORALLY HETEROGENEOUS DATA

We evaluate CATS in a class-incremental FCL system under latency constraints. The system consists of one edge server and multiple devices. In each training round, the server selects a subset of devices to perform local training and upload their model updates. CATS schedules devices using a time-varying combination of temporal drift and collective divergence: temporal drift identifies devices whose newly arrived data contain substantial distributional changes, whereas collective divergence evaluates the global representativeness of the selected device group.

We use the CIFAR-100 dataset to evaluate the continual learning performance, and consider an FCL system with 30 devices and one edge server. ResNet-18 is adopted as the learning model. The model is initially trained on class indexes 0–29, and new classes are then introduced gradually, after which classes 30–39 and 40–44 are introduced in successive stages. At each incremental stage, 12 devices receive samples from the newly introduced classes, and the per-round latency budget is set to 30s.

For comparison, we consider several scheduling baselines. Random scheduling selects devices randomly in each round. Best Channel prioritizes devices with favorable wireless channel conditions [10]. Power-of-Choice selects devices with

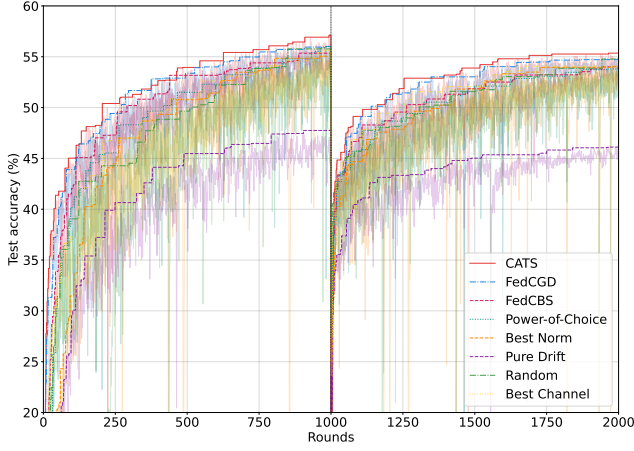


Fig. 4. Performance of CATS for learning from spatio-temporally heterogeneous data.

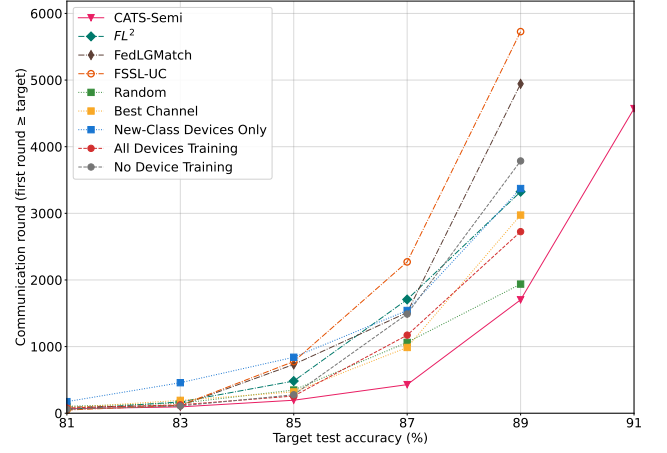
larger local losses, while Best Norm gives priority to devices with larger local update norms [11]. FedCBS addresses spatial data heterogeneity by selecting devices based on class imbalance in their local datasets [12]. Our previous work, FedCGD, schedules device groups solely according to collective divergence [8]. We also include Pure Drift as an ablation baseline that selects devices solely based on temporal drift.

Fig. 4 shows the test accuracy of different scheduling schemes during the continual learning process. The lightly shaded curves represent the round-by-round test accuracy, while the solid curves indicate the best test accuracy achieved up to each communication round. The arrival of new classes causes a temporary accuracy drop for all methods, reflecting the need to adapt to the changed data distribution. Random scheduling and Best Channel cannot fully capture the learning value of newly arrived data, while FedCBS and FedCGD mainly emphasize distributional representativeness under non-i.i.d. data. Pure Drift can quickly respond to data changes, but may over-emphasize new data and increase the risk of forgetting. By jointly considering temporal drift and collective divergence, CATS selects device groups that capture newly emerging knowledge while maintaining global representativeness. Consequently, it recovers more rapidly after each distribution change and maintains more stable performance across continual learning stages.

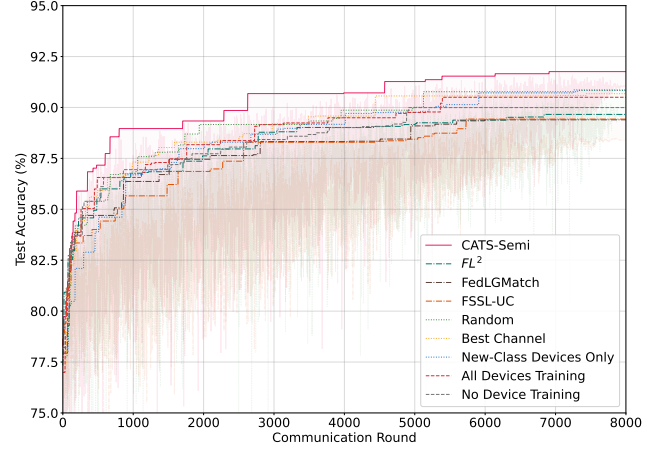
V. CASE STUDY II: LEARNING FROM UNLABELED DATA

We further evaluate CATS-Semi in a latency-constrained semi-supervised FCL system. The objective is to exploit device-side unlabeled data under limited server-side annotations while avoiding unreliable pseudo-labels that may mislead continual model updates.

The edge server first initializes the global model using its limited labeled data and broadcasts the model to the devices. Candidate devices generate pseudo-labels for their local unlabeled samples and report lightweight statistics describing pseudo-label quantity and reliability. Based on these statistics, the server then schedules a subset of devices for local training. Only the scheduled devices train on the retained



(a) Communication rounds required to reach different target accuracies.



(b) Test accuracy over communication rounds.

Fig. 5. Performance of CATS-Semi for learning from massive unlabeled data.

pseudo-labeled samples and upload their model updates. After aggregation, the server further fine-tunes the global model using its labeled reference data.

The system consists of 30 devices and one edge server. A convolutional neural network is learned over the SVHN dataset. The dataset contains 10 classes, with class indexes 0–5 used as the initial classes. At the incremental stage, the edge server maintains 1,000 labeled samples from previously learned classes and 100 labeled samples from newly introduced classes. Each device holds 800 unlabeled samples, while only 12 devices contain data from the newly introduced classes. The per-round latency budget is set to 1s. This setting represents a label-scarce environment in which devices continuously generate abundant unlabeled data, while only limited labeled samples are available at the edge server.

For comparison, we consider several baselines under the same semi-supervised continual learning setting. New-Class Devices Only selects only devices containing newly arrived classes, while All Devices Training lets all available devices participate in training. No Device Training only uses the limited labeled data at the edge server without involving device-side unlabeled data. We further compare CATS-Semi with representative federated semi-supervised learning methods, including FL^2 for few-label learning [13], FedLGMATCH

for local-global pseudo-label matching [14], and FSSL-UC for uncertainty and consistency aware training [15].

Fig. 5 shows the performance of different methods under massive unlabeled data. As shown in Fig. 5(a), CATS-Semi requires fewer communication rounds to reach different target accuracies, with a more pronounced advantage at higher accuracy targets. This indicates that simply using more unlabeled data, selecting only devices with new classes, or relying only on favorable channel conditions is not sufficient for label-efficient continual learning. CATS-Semi instead prioritizes devices that offer a favorable balance between pseudo-label quantity and reliability. As shown in Fig. 5(b), CATS-Semi achieves higher test accuracy during continual training, indicating that it can better exploit unlabeled data and improve model generalization under resource constraints. As all methods operate under the same per-round latency budget, the reduction in communication rounds directly translates into lower adaptation latency. Overall, these results demonstrate that balancing pseudo-label quantity and reliability improves both label efficiency and the timeliness of FCL.

VI. CONCLUSION AND OUTLOOK

This article presented CATS, a device scheduling framework based on collective and temporal synergy for timely FCL at the wireless edge. Rather than evaluating devices independently, CATS characterizes the group-level learning value of scheduled devices and jointly considers learning contribution and resource feasibility. Under spatio-temporal data heterogeneity, CATS dynamically balances temporal drift and collective divergence to accelerate adaptation to newly arrived data while maintaining global representativeness. Its semi-supervised extension, CATS-Semi, further balances pseudo-label quantity and reliability across newly introduced and previously learned classes, enabling effective use of unlabeled data while mitigating confirmation bias and error accumulation. The case studies demonstrate that CATS accelerates model convergence and stabilizes continual learning, while CATS-Semi reduces the number of communication rounds required to reach target performance and improves label-efficient adaptation.

Several future directions deserve further investigation. First, mobility-aware FCL is important for dynamic edge systems with vehicles, drones, and mobile robots, where device mobility changes both wireless channels and local data distributions. The interplay among mobility, data drift, and communication opportunities should be jointly modeled. Second, active learning can be integrated with FCL. Since labels are scarce and expensive, the system may actively select the most valuable or uncertain samples for annotation, thereby reducing labeling costs and improving new-class learning. Third, resource-efficient continual learning requires effective training-triggering mechanisms. Instead of updating models at fixed intervals, future systems may trigger training only when significant data drift or performance degradation is detected. Designing reliable drift detection and training-triggering policies under privacy and resource constraints remains an important open problem.

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